

MODULE 3

Vector Differential Calculus

Mathematics-I — MA101

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Syllabus Coverage

Topic
Derivatives of vector valued functions; vector equations of curves; tangents of a curve
Gradient and directional derivative of a scalar field
Divergence of a vector field
Curl of a vector field
Line integrals and double integrals

Chapter 1: Vector-Valued Functions and Parametric Curves

The application of vector calculus to geometry is called differential geometry. Bodies that move in space trace out paths that can be represented by curves C . This motivates the parametric representation of a curve C , using a parameter t (which may denote time or some other quantity).

1.1 Parametric Representation of a Curve

A curve C in space is represented parametrically by the position vector:

$$r(t) = [x(t), y(t), z(t)] = x(t) i + y(t) j + z(t) k$$

Here t is the parameter and x, y, z are the usual Cartesian coordinates. Each value $t = t_0$ corresponds to a point of C with position vector $r(t_0)$, whose coordinates are $x(t_0), y(t_0), z(t_0)$.

Note: Parametric form (1) treats x, y, z symmetrically as independent variables and induces an orientation on C — the sense of increasing t is the positive sense; the sense of decreasing t is the negative sense.

1.2 Straight Line, Plane, Twisted and Simple Curves

A straight line through (x_1, y_1, z_1) with direction ratios l, m, n has the symmetric (parametric) form:

$$(x - x_1)/l = (y - y_1)/m = (z - z_1)/n$$

Term	Meaning
Plane curve	A curve that lies entirely in a plane in space
Twisted curve	A curve that is not planar
Simple curve	A curve without multiple points (does not intersect or touch itself); e.g. circle, helix
Arc	The portion of a curve between any two of its points

1.3 Tangent to a Curve

The tangent to a simple curve C at a point P is the limiting position of a line L through P and a nearby point Q of C as Q approaches P along C . If P and Q correspond to parameters t and $t + \Delta t$, a vector along L is:

$$(1/\Delta t)[r(t + \Delta t) - r(t)]$$

Letting $\Delta t \rightarrow 0$ gives the derivative $r'(t)$, tangent to C at P . The unit tangent vector is:

$$u = (1/|r'|) r'$$

The tangent line itself, with parameter w , is given by:

$$q(w) = r + w r'$$

This is the sum of the position vector r of P and a multiple of the tangent vector r' of C at P .

Chapter 2: Length of a Curve and the Unit Tangent

2.1 Arc Length

If $r(t)$ has a continuous derivative, the length of the curve C between $t = a$ and $t = b$ — assuming C is rectifiable — is given by the integral:

$$l = \int[a \rightarrow b] \sqrt{r' \cdot r'} dt$$

Worked Example — dr/ds is a Unit Tangent of C

If s is the arc length of a curve $C: r(t)$, prove that dr/ds is a unit tangent of C .

Solution:

We have $s(t) = \int[a \rightarrow t] \sqrt{r' \cdot r'} d\tilde{t}$, where $r' = dr/dt$. By the Fundamental Theorem of Calculus:

$$ds/dt = \sqrt{dr/dt \cdot dr/dt} = |dr/dt| \quad \dots (A)$$

By the chain rule, $dr/dt = (dr/ds)(ds/dt)$. Substituting into the dot product $dr/dt \cdot dr/dt$ and using (A):

$$(ds/dt)^2 = |dr/ds|^2 (ds/dt)^2 \Rightarrow |dr/ds| = 1$$

Moreover, since $dr/dt = (dr/ds)(ds/dt)$, the vector dr/ds is parallel to dr/dt . Hence dr/ds is a unit vector that is tangent to C — that is, dr/ds is a unit tangent of C .

Chapter 3: Directional Derivatives and Gradient of a Scalar Field

3.1 Gradient of a Scalar Field

Let $\nabla = i \partial/\partial x + j \partial/\partial y + k \partial/\partial z$ be the vector differential (del) operator. The gradient of a scalar function f is defined as:

$$\text{grad } f = \nabla f = (\partial f/\partial x) i + (\partial f/\partial y) j + (\partial f/\partial z) k$$

Worked Example — Gradient of $f(x, y, z) = xyz$

Find the gradient of $f(x, y, z) = xyz$ at the point $P(1, 2, 3)$.

Solution:

By definition, $\text{grad } f = \partial f/\partial x i + \partial f/\partial y j + \partial f/\partial z k = yz i + xz j + xy k$.

(The value at P is not required further here — the general gradient field is what is used in later examples.)

3.2 Directional Derivative

Let $f(x, y, z)$ be a scalar function. Its directional derivative at P in the direction of a unit vector b is the rate of change of f at P in that direction:

$$D_b f = df/ds = \lim_{s \rightarrow 0} [f(Q) - f(P)] / s$$

where Q is a variable point on the ray $C : r(s) = p_0 + s b$, and s is the distance between P and Q . Using the chain rule for $f(x(s), y(s), z(s))$:

$$D_b f = (\text{grad } f) \cdot dr/ds$$

Since $dr/ds = b$ for the ray $r(s) = p_0 + s b$, this gives:

$$D_b f = b \cdot (\text{grad } f)$$

More generally, for any vector $a \neq 0$ (not necessarily a unit vector), the directional derivative of f in the direction of a is:

$$D_a f = (1/|a|) (a \cdot \text{grad } f)$$

Worked Example — Directional Derivative of $f(x, y) = e^x \cos y$

Find the directional derivative of $f(x, y) = e^x \cos y$ at the point $P(2, \pi, 0)$ in the direction of $a = 2i + 3j$.

Solution:

$\text{grad } f = (\partial f/\partial x) i + (\partial f/\partial y) j = e^x \cos y i - e^x \sin y j$.

At $(2, \pi, 0)$: $[\nabla f] = e^2 \cos \pi i + (-e^2 \sin \pi) j = -e^2 i$.

$$D_a f = [a/|a|] \cdot \nabla f = [(2i + 3j)/\sqrt{13}] \cdot (-e^2 i) = -2e^2/\sqrt{13}$$

3.3 Gradient is Normal to a Level Surface

Theorem: Let f be a differentiable scalar function representing a surface S : $f(x, y, z) = c$ (constant). If $\text{grad } f$ at a point P of S is not the zero vector, then $\text{grad } f$ is a normal vector of S at P .

Sketch of proof: For any curve C : $r(t)$ on S , $f(x(t), y(t), z(t)) = c$, so $d f/dt = 0$. By the chain rule this gives $(\text{grad } f) \cdot r' = 0$. Since r' is tangent to S , $\text{grad } f$ is perpendicular to every tangent direction at P — hence $\text{grad } f$ is normal to S at P .

3.4 Useful Results Involving the Gradient

Identity	Statement
Product rule	$\nabla(fg) = f \nabla g + g \nabla f$
Laplacian of a product	$\nabla^2(fg) = g \nabla^2 f + f \nabla^2 g + 2(\nabla f \cdot \nabla g)$
Quotient rule	$\nabla(f/g) = (1/g^2)(g \nabla f - f \nabla g)$

Note: These identities are proved by expanding the gradient/Laplacian operator component-wise and applying the product rule of ordinary differentiation to each Cartesian component.

Chapter 4: Divergence of a Vector Field

4.1 Definition

For a vector function $v(x, y, z) = v_1 i + v_2 j + v_3 k$, the divergence is the scalar function:

$$\text{div } v = \nabla \cdot v = \partial v_1 / \partial x + \partial v_2 / \partial y + \partial v_3 / \partial z$$

Worked Example — Divergence of a Given Field

Find the divergence of $v = 2x^2z i - xy^2 j + 3yz^2 k$.

Solution:

$$\text{div } v = \partial / \partial x (2x^2z) + \partial / \partial y (-xy^2) + \partial / \partial z (3yz^2) = 4xz - 2xy + 6yz.$$

Worked Example — Divergence of an Exponential Field

Find the divergence of $v = e^{-x} i + e^{-x} y j + 2z \sinh x k$.

Solution:

$$\text{div } v = \partial / \partial x (e^{-x}) + \partial / \partial y (e^{-x} y) + \partial / \partial z (2z \sinh x) = -e^{-x} + e^{-x} + 2 \sinh x = 2 \sinh x = e^x - e^{-x}.$$

4.2 Physical Significance

The divergence of a vector field at a point P represents the volume density of outward flux at that point — it measures the net rate at which the field ‘spreads out’ from P . If the flow is incompressible, the divergence vanishes:

$$\text{div } v = 0 \text{ (incompressible flow)}$$

4.3 Useful Results Involving Divergence

Identity	Statement
Scalar multiple	$\operatorname{div}(k \mathbf{v}) = k \operatorname{div} \mathbf{v}$, for constant k
Product with a scalar field	$\operatorname{div}(f \mathbf{v}) = f \operatorname{div} \mathbf{v} + \mathbf{v} \cdot \nabla f$
Divergence of $f \nabla g$	$\operatorname{div}(f \nabla g) = f \nabla^2 g + (\nabla f \cdot \nabla g)$

Practice Problem — Laplacian of xy/z

Find the value of $\nabla^2 f$ if $f = xy/z$.

Solution:

$$\partial f / \partial x = y/z, \partial f / \partial y = x/z, \partial f / \partial z = -xy/z^2.$$

$$\partial^2 f / \partial x^2 = 0, \partial^2 f / \partial y^2 = 0, \partial^2 f / \partial z^2 = 2xy/z^3.$$

$$\nabla^2 f = 0 + 0 + 2xy/z^3 = 2xy/z^3$$

Chapter 5: Curl of a Vector Field

5.1 Definition

The curl operator, denoted $\nabla \times$, acts on a vector field and returns another vector field. If $\mathbf{A} = ax \mathbf{i} + ay \mathbf{j} + az \mathbf{k}$, then $\operatorname{curl} \mathbf{A} = \nabla \times \mathbf{A}$, best remembered in determinant form:

$$\operatorname{curl} \mathbf{A} = \mathbf{i} (\partial az / \partial y - \partial ay / \partial z) + \mathbf{j} (\partial ax / \partial z - \partial az / \partial x) + \mathbf{k} (\partial ay / \partial x - \partial ax / \partial y)$$

Note: If $\nabla \times \mathbf{A} = \mathbf{0}$ throughout a region, $\mathbf{A}$ is said to be irrotational there.

Worked Example — Curl of a Vector Field

If $\mathbf{A} = (y^4 - x^2 z^2) \mathbf{i} + (x^2 + y^2) \mathbf{j} - x^2 y z \mathbf{k}$, determine $\operatorname{curl} \mathbf{A}$ at the point $(1, 3, -2)$.

Solution:

Expanding the determinant gives $\operatorname{curl} \mathbf{A} = \mathbf{i}(-x^2 z) - \mathbf{j}(-2xyz + 2x^2 z) + \mathbf{k}(2x - 4y^3) = -x^2 z \mathbf{i} - (2x^2 z - 2xyz) \mathbf{j} + (2x - 4y^3) \mathbf{k}$.

$$\text{At } (1, 3, -2): \operatorname{curl} \mathbf{A} = 2 \mathbf{i} - 8 \mathbf{j} - 106 \mathbf{k}$$

5.2 Physical Significance

The curl of a vector field at a point P gives the (local) angular velocity of the field there. A vector field is said to be irrotational if $\operatorname{curl} \mathbf{v} = \mathbf{0}$ at every point; otherwise it is rotational.

5.3 Useful Results on Curl

Identity	Statement
Linearity	$\operatorname{curl}(\mathbf{u} + \mathbf{v}) = \operatorname{curl} \mathbf{u} + \operatorname{curl} \mathbf{v}$
Divergence of a curl	$\operatorname{div}(\operatorname{curl} \mathbf{v}) = 0$, for any twice-differentiable $\mathbf{v}$

Note: $\text{div}(\text{curl } \mathbf{v}) = 0$ follows because each term of the expansion involves a mixed second partial derivative that cancels with an identical term of opposite sign, using the equality of mixed partial derivatives ($\partial^2/\partial x \partial y = \partial^2/\partial y \partial x$, etc.).

Chapter 6: Line Integrals

6.1 Introduction

Let C be a curve with parametric representation $\mathbf{r}(t) = [x(t), y(t), z(t)] = x(t)\mathbf{i} + y(t)\mathbf{j} + z(t)\mathbf{k}$, $a \leq t \leq b$. C is called the path of integration; $A: \mathbf{r}(a)$ is its initial point and $B: \mathbf{r}(b)$ its terminal point. The direction of increasing t is the positive direction on C . If A and B coincide, C is a closed path.

6.2 Physical Applications

Application
Mass of a wire
Centre of mass and moments of inertia of a wire
Work done by a force on an object moving in a vector field
Magnetic field around a conductor (Ampère's Law)
Voltage generated in a loop (Faraday's Law of magnetic induction)

6.3 General Properties of the Line Integral

Property	Statement
Scalar multiple	$\int_C k \mathbf{F}(\mathbf{r}) \cdot d\mathbf{r} = k \int_C \mathbf{F}(\mathbf{r}) \cdot d\mathbf{r}$, k constant
Linearity	$\int_C (a_1 \mathbf{G}_1 + a_2 \mathbf{G}_2 + \dots + a_n \mathbf{G}_n) \cdot d\mathbf{r} = a_1 \int_C \mathbf{G}_1 \cdot d\mathbf{r} + a_2 \int_C \mathbf{G}_2 \cdot d\mathbf{r} + \dots + a_n \int_C \mathbf{G}_n \cdot d\mathbf{r}$
Additivity over sub-paths	$\int_C \mathbf{F} \cdot d\mathbf{r} = \int_{C_1} \mathbf{F} \cdot d\mathbf{r} + \int_{C_2} \mathbf{F} \cdot d\mathbf{r} + \dots + \int_{C_n} \mathbf{F} \cdot d\mathbf{r}$, $C = C_1 + C_2 + \dots + C_n$
Reversal of direction	$\int [a \rightarrow b] \mathbf{F} \cdot d\mathbf{r} = - \int [b \rightarrow a] \mathbf{F} \cdot d\mathbf{r}$

6.4 Line Integral of a Scalar Field

If a scalar field V exists at all points of a curve C , the line integral of V along C from A to B is $\int_C V d\mathbf{r}$, obtained by expressing V and $d\mathbf{r}$ in terms of x, y, z along the parametrisation of C .

Worked Example — Line Integral of a Scalar Field

If $V = xy + y^2z$, evaluate $\int_C V d\mathbf{r}$ along the curve $C: x = t^2, y = 2t, z = t + 5$, between $A(0, 0, 5)$ and $B(4, 4, 7)$.

Solution:

Along C , $V = (t^2)(2t) + (4t^2)(t + 5) = 6t^3 + 20t^2$, and $d\mathbf{r} = 2t dt \mathbf{i} + 2 dt \mathbf{j} + dt \mathbf{k}$. A corresponds to $t = 0$ and B to $t = 2$, so:

$$\int_C \mathbf{V} \cdot d\mathbf{r} = \int_{[0 \rightarrow 2]} (6t^3 + 20t^2)(2t \mathbf{i} + 2t \mathbf{j} + \mathbf{k}) dt = (8/15)(444 \mathbf{i} + 290 \mathbf{j} + 145 \mathbf{k})$$

6.5 Line Integral of a Vector Field

If a vector field $\mathbf{F} = F_x \mathbf{i} + F_y \mathbf{j} + F_z \mathbf{k}$ exists at all points of C , its line integral from A to B is:

$$\int_C \mathbf{F} \cdot d\mathbf{r} = \int_C F_x dx + \int_C F_y dy + \int_C F_z dz$$

Since $\mathbf{F} \cdot d\mathbf{r}$ is a scalar product, the line integral is a scalar. If $\mathbf{F}$ is a force field, $\int_C \mathbf{F} \cdot d\mathbf{r}$ represents the work done in moving a unit particle along C from A to B .

Worked Example — Line Integral of a Vector Field

If $\mathbf{F} = x^2y \mathbf{i} + xz \mathbf{j} - 2yz \mathbf{k}$, evaluate $\int_C \mathbf{F} \cdot d\mathbf{r}$ between $A(0, 0, 0)$ and $B(4, 2, 1)$ along the curve $x = 4t$, $y = 2t^2$, $z = t^3$.

Solution:

Along C , $\mathbf{F} = 32t^4 \mathbf{i} + 4t^4 \mathbf{j} - 4t^5 \mathbf{k}$ and $d\mathbf{r} = 4 dt \mathbf{i} + 4t dt \mathbf{j} + 3t^2 dt \mathbf{k}$, with $t = 0$ at A and $t = 1$ at B . Then:

$$\int_C \mathbf{F} \cdot d\mathbf{r} = \int_{[0 \rightarrow 1]} (128t^4 + 16t^5 - 12t^7) dt = 128/5 + 8/3 - 3/2 = 803/30 \approx 26.77$$

6.6 Path Dependence of a Line Integral

For $\mathbf{F} = x^2y \mathbf{i} + 2yz \mathbf{j} + 3z^2x \mathbf{k}$, evaluate $\int_C \mathbf{F} \cdot d\mathbf{r}$ between $A(0, 0, 0)$ and $B(1, 2, 3)$: (a) along the straight-line segments $C_1: (0,0,0) \rightarrow (1,0,0)$, $C_2: (1,0,0) \rightarrow (1,2,0)$, $C_3: (1,2,0) \rightarrow (1,2,3)$; (b) along the single straight line C_4 joining $(0,0,0)$ to $(1,2,3)$.

Solution:

$\mathbf{F} \cdot d\mathbf{r} = x^2y dx + 2yz dy + 3z^2x dz$. On C_1 ($y = z = 0$) and C_2 ($x = 1, z = 0$) every term vanishes, giving zero on both. On C_3 ($x = 1, y = 2$): $\int_{C_3} \mathbf{F} \cdot d\mathbf{r} = \int_{[0 \rightarrow 3]} 3z^2 dz = 27$. Summing, $\int_{(C_1+C_2+C_3)} \mathbf{F} \cdot d\mathbf{r} = 27$.

On C_4 : $x = t, y = 2t, z = 3t$ ($0 \leq t \leq 1$), so $\mathbf{F} = 2t^3 \mathbf{i} + 12t^2 \mathbf{j} + 27t^3 \mathbf{k}$ and $d\mathbf{r} = (dt \mathbf{i} + 2dt \mathbf{j} + 3dt \mathbf{k})$. Then:

$$\int_{C_4} \mathbf{F} \cdot d\mathbf{r} = \int_{[0 \rightarrow 1]} (2t^3 + 24t^2 + 81t^3) dt = 83/4 + 8 = 115/4 = 28.75$$

Note: The value of the line integral (27 via $C_1+C_2+C_3$, versus 28.75 via C_4) depends on the path taken between the same two endpoints — this $\mathbf{F}$ is not a conservative (path-independent) field.

6.7 Line Integrals Independent of Path

A line integral $\int_C \mathbf{F}(\mathbf{r}) \cdot d\mathbf{r} = \int_C (F_1 dx + F_2 dy + F_3 dz)$ is independent of path in a domain D if, for every pair of endpoints A and B in D , it has the same value along every path in D joining A to B .

A domain D is simply connected if every closed curve in D can be shrunk continuously to a point of D without leaving D .

Theorem	Statement
Theorem 3 (necessary condition)	If F_1, F_2, F_3 are continuous with continuous first partials in D and $\int_C (F_1 dx + F_2 dy + F_3 dz)$ is independent of path in D , then $\text{curl } \mathbf{F} = \mathbf{0}$ in D — i.e. $\partial F_3 / \partial y = \partial F_2 / \partial z$, $\partial F_1 / \partial z = \partial F_3 / \partial x$, $\partial F_2 / \partial x = \partial F_1 / \partial y$.

Theorem	Statement
Theorem 4 (sufficient condition)	If F_1, F_2, F_3 are continuous with continuous first partials in D , $\text{curl } F = 0$ in D , and D is simply connected, then $\int_C (F_1 dx + F_2 dy + F_3 dz)$ is independent of path in D .

Chapter 7: Double Integrals

7.1 Definition

Let $f(x, y)$ be a function defined on a region R of the plane. Subdivide R into small rectangular subregions, the k -th of dimensions $\Delta x_k, \Delta y_k$ and area $\Delta A_k = \Delta x_k \Delta y_k$. Choosing an arbitrary point (x_k, y_k) in each subregion, form the Riemann sum:

$$J_n = \sum_{k=1}^n f(x_k, y_k) \Delta A_k$$

If f is continuous on R and R is bounded by finitely many smooth curves, the sequence $(J_1, J_2, \dots)$ converges to a limit independent of the choice of subdivisions and sample points. This limit is the double integral of f over R :

$$\iint_R f(x, y) dx dy \quad \text{or} \quad \iint_R f(x, y) dA$$

7.2 Applications of Double Integrals

Application
Area of a plane region
Volume beneath a surface (e.g. an elliptic paraboloid)
Average value of a function over a region
Surface integrals
Mass, centre of mass, and moments of a lamina

7.3 Properties of the Double Integral

Property	Statement
Linearity (sum)	$\iint_A [f(x, y) + g(x, y)] dA = \iint_A f dA + \iint_A g dA$
Linearity (scalar)	$\iint_A c f(x, y) dA = c \iint_A f(x, y) dA$
Comparison	If $f(x, y) \geq g(x, y)$ on R , then $\iint_A f dA \geq \iint_A g dA$

Worked Example — Volume Beneath a Surface

Find the volume of the region beneath $z = 4x^2 + 9y^2$ over the rectangle with vertices $(0,0)$, $(3,0)$, $(3,2)$, $(0,2)$.

Solution:

The volume beneath $z = f(x, y)$ (> 0) and above region R in the xy -plane is $V = \iint_R f(x, y) \, dx \, dy$. Integrating first over x from 0 to 3:

$$V = \int_{y=0 \rightarrow 2} \int_{x=0 \rightarrow 3} (4x^2 + 9y^2) \, dx \, dy = \int_{y=0 \rightarrow 2} (36 + 27y^2) \, dy = [36y + 9y^3] \text{ from } 0 \text{ to } 2$$

$$V = 72 + 72 = 144$$

Quick Reference — All Key Formulae (Module 3)

Curves and Tangents

$$r(t) = x(t)i + y(t)j + z(t)k; \text{ tangent line: } q(w) = r + w r'; \text{ unit tangent: } u = r'/|r'|$$

Arc Length

$$l = \int_{a \rightarrow b} |v(r') \cdot r'| \, dt; \text{ } dr/ds \text{ is a unit tangent of } C$$

Gradient and Directional Derivative

$$\text{grad } f = \nabla f = f_x i + f_y j + f_z k; \text{ } D_a f = (1/|a|)(a \cdot \nabla f); \text{ } \nabla f \text{ is normal to } f = c$$

Divergence

$$\text{div } v = \nabla \cdot v = \partial v_1 / \partial x + \partial v_2 / \partial y + \partial v_3 / \partial z; \text{ } \text{div } v = 0 \text{ for incompressible flow}$$

Curl

$$\text{curl } A = \nabla \times A \text{ (determinant form); } \text{curl } v = 0 \Leftrightarrow v \text{ irrotational; } \text{div}(\text{curl } v) = 0$$

Line Integrals

$$\int_C F \cdot dr = \int_C F_x dx + F_y dy + F_z dz; \text{ independent of path in simply connected } D \Leftrightarrow \text{curl } F = 0$$

Double Integrals

$$\iint_R f(x, y) \, dA = \lim \sum f(x_k, y_k) \Delta A_k; \text{ used for area, volume, mass, and averages}$$

— End of Module 3 Notes —